Computers and Education: Artificial Intelligence 8 (2025) 100416

Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence

Understanding AI adoption among secondary education teachers: A pls-sem approach

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ARTICLE INFO

Keywords:
AI adoption
Secondary school teachers
AI knowledge
Data literacy
Partial least squares structural equation modelling (PLS-SEM)

ABSTRACT

This study investigates the factors influencing the adoption of Artificial Intelligence (AI) by secondary school teachers in Catalonia. Using a Partial Least Squares Structural Equation Modelling (PLS-SEM) methodology, a conceptual model was analyzed that includes AI perception, AI knowledge, General data use, Applied data use, and STEM training as predictors of AI adoption. The results reveal that AI knowledge ($\beta = .482$, $p < .001$) and General data use ($\beta = .288$, $p = .001$) are the most significant and positive predictors of AI adoption. In contrast, AI perception shows a weak but statistically significant negative relationship ($\beta = -.105$, $p = .022$), while applied data use and STEM training do not present a significant direct effect. The model explains 30.5 % of the variance in AI adoption. These findings suggest that developing specific knowledge on how to use AI for content creation and competence in general data use is crucial to fostering AI adoption among secondary school teachers in the Catalan context. In addition, this explorative work provides the research community with evidence that key Data Literacy competencies significantly shape AI adoption.

1. Introduction

The burgeoning integration of artificial intelligence (AI) into educational paradigms is reshaping pedagogical practices and promising to deliver highly personalized learning experiences. Driven by advancements in machine learning, natural language processing, and intelligent tutoring systems, AI facilitates an advanced analysis of student learning data, enabling the creation of tailored pedagogical strategies. Adaptive learning platforms, increasingly prevalent, exemplify this shift, leveraging AI to dynamically adjust educational content to individual learning styles and paces, thereby enabling deeper comprehension and enhanced problem-solving skills. Beyond personalization, AI also catalyses collaborative learning, fostering meaningful interactions between students and educators and promoting constructive social learning environments (Xu et al., 2022). This hybrid approach, merging personalized instruction with the dynamism of collaborative activities, underscores AI's potential to enrich the educational experience.

However, the effective deployment of AI hinges on recognizing the key role of educators. AI systems should serve as augmentation tools for teachers, whose professional judgment, pedagogical expertise, and capacity to build meaningful student relationships remain paramount (Holstein & Aleven, 2022). Maintaining teacher autonomy in pedagogical decision-making is crucial for ethical and responsible AI implementation (Echeverría et al., 2023; Lawrence et al., 2024). Ethical considerations are central to AI's integration. Data privacy, security, and the imperative to prevent the perpetuation of existing inequalities necessitate careful attention (Holstein & Aleven, 2022; Li et al., 2022; Moon et al., 2022; Thomas et al., 2024, pp. 404–415). Teachers' trust in AI-based educational technologies is built upon the transparency of AI decision-making processes. Teachers must understand, at least at a basic level, how AI algorithms work to feel comfortable using them in the classroom. This requires a foundation in STEM, particularly in understanding data and algorithms. Factors such as self-efficacy in using AI tools, the comparison between AI recommendations and their own pedagogical decisions, anxieties related to AI, and the perceived benefits in learning improvement and work efficiency influence teacher confidence (Nazaretsky et al., 2022).

Studies indicate that AI perception factors, especially perceived benefits, ease of use, and usefulness, consistently relate to the adoption

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https://doi.org/10.1016/j.caeai.2025.100416
Received 12 March 2025; Received in revised form 5 May 2025; Accepted 6 May 2025
Available online 13 May 2025
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of AI technologies. Viberg et al. (2024) observed that perceptions of benefit account for nearly 50 % of the variance in trust toward AI-EdTech, while Choi, Jang, and Kim (2023), Sun et al. (2024), and Li (2024) reported significant positive effects of ease of use and usefulness. Moreover, Aguilar-Cruz and Salas-Pilco (2025) and Du, H. et al. (2024) underscore the critical role of perceptions in AI adoption. The former specifically investigates Colombian teachers' views on AI in education, encompassing perceived benefits, challenges, and concerns, notably highlighting ethical issues around transparency and data privacy and suggesting that these perceptions can reveal hurdles in AI's technological integration, thus influencing adoption. This highlights that teachers' perceptions can reveal issues related to the technological integration processes of AI, which supports the relevance of AI perception as a factor influencing adoption.

Regarding AI knowledge, Du, H. et al. (2024) focus on "AI Literacy", defined as a set of competencies that enable individuals to critically evaluate, communicate effectively about, and utilize AI as a tool. AI literacy encompasses the knowledge and understanding of AI concepts and applications. This research found that AI literacy directly influences self-efficacy in learning AI, perceptions of the use of AI for social good, and awareness of AI ethics. Given that self-efficacy (confidence in one's ability) and attitudes (perceptions) predict behavioural intentions to use or learn technologies, establishing these theoretical links justifies the inclusion of AI Knowledge as a relevant factor. It also found that teachers' age and specialization could influence their awareness of AI ethics. The results suggest that individuals with a background in information systems (which could be related to STEM) might better understand ethical issues. Therefore, STEM may be a determining factor to analyze.

Combined frameworks such as the Technology Acceptance Model (TAM), UTAUT2, and TPACK are used to explain behavioural intentions, with one UTAUT2 study attributing 75.2 % of the variance to this factor (Cabero-Almenara et al., 2024). AI knowledge plays a positive but varied role, and measures of general data use typically emerge within broader technical mastery (Viberg et al., 2024). Sun et al. (2024) include pre-service STEM teachers and high school science teachers and suggest that a STEM background may favour AI integration, though direct comparisons remain limited. In addition, contextual factors—organizational support, ethical and workload considerations, cultural dimensions, and parental involvement—further shape the overall influence of these predictors in educational settings (Bayaga, 2024). Furthermore, Hwang and Chien (2024) and Hwang et al. (2021) mention that the perceptions of stakeholders, including teachers, determine the success of new technology-based educational modes. The authors suggest that theoretical models such as the TAM or its extensions can be used to find the factors affecting acceptance or perceptions. Traditional TAM includes constructs such as perceived usefulness and perceived ease of use, among others, which are aspects of perception. The present study is contextualized by the fact that the perception of benefits, ease of use, and usefulness are consistently related to technology adoption, which aligns with TAM and other cited research.

A successful adoption of AI requires teachers' and students' predispositions toward its use. Therefore, building AI literacy is fundamental for both (Ng et al., 2023). Teachers must acquire the necessary skills to integrate AI into their pedagogical practices effectively and ethically. Professional development programs play a central role and should consider combining AI concepts with hands-on experiences, exploring ethical dilemmas, and fostering collaborative learning among teachers. The most effective training strategies include (Le & Perret, 2022; Lee et al., 2024).

- Practical experiences where teachers use AI tools from the students' perspective.
- Contextualization of AI concepts in real-world educational applications.
- Analysis of ethical aspects and potential consequences of using AI in education.
- Promotion of collaborative learning so that teachers share experiences and strategies.

The integration of AI demands a revised and updated teacher profile that goes beyond mere technical competencies. Frameworks such as DigCompEdu, the P21 framework (Ng et al., 2023), and the recent UNESCO AI competency framework for teachers (Miao & Cukurova, 2024) provide essential guidance in defining the competencies required for educators to navigate this evolving landscape. These frameworks emphasize the ability to design AI-enhanced learning experiences, evaluate results using AI, and adapt instruction to diverse student needs. Specifically, DigCompEdu underscores the necessity for educators to develop digital competence, including an understanding of AI's potential and limitations, the ability to integrate AI into teaching strategies, and the capacity to promote critical thinking and digital literacy. The UNESCO AI competency framework for teachers and the P21 framework further detail the essential capabilities for educators, including the ability to design learning experiences, evaluate results with AI, and adapt to the needs of students.

The adoption of AI in education requires a holistic approach encompassing technological, pedagogical, ethical, and social aspects. Cultivating teachers' trust in AI, developing their AI literacy, and promoting updated teaching competencies are crucial for a successful transformation. Future research should focus on developing models that incorporate the active participation of teachers in the design of AI tools.

This study explores the factors influencing the adoption of artificial intelligence (AI) by secondary school teachers in Catalonia, with a special emphasis on its combination with data literacy. The context of this research is the increasing integration of AI into educational paradigms, which is transforming pedagogical practices and promises to offer highly personalized learning experiences. However, the effective implementation of AI depends on the recognition of the key role of educators. This work is situated in the need to understand how data literacy-related competencies and AI knowledge, together with perceptions about AI and other sociodemographic variables, influence the use of AI technologies for educational purposes.

2. Methodology

2.1. Research design and statistical approach

The Partial Least Squares Structural Equation Modelling (PLS-SEM) method was selected to predict the level of AI Adoption. As a variance-based multivariate technique, PLS-SEM analyzes complex relationships involving latent variables and their corresponding indicators. This approach is increasingly recognized for exploratory research and theory development, particularly when examining emerging fields with limited theoretical consensus Hair & Alamer, 2022).

Thus, the selection criteria for developing PLS-SEM rather than CB-SEM (Covariance-based Structural Equation Modelling) in this research stem from the study's exploratory nature and the results of the skewness and kurtosis tests, which revealed significant deviations from the normal distribution. Therefore, while CB-SEM requires multivariate normality, PLS-SEM accepts non-normal distributions (Hair et al., 2022).

In addition, a review of prior research reveals that models such as the Value Added Model (VAM), Theory of Planned Behaviour (TPB), Unified Theory of Acceptance and Use of Technology (UTAUT), and Technology Acceptance Model (TAM) have been widely used to examine AI adoption in educational settings (Wang et al., 2021). However, empirical validation of these frameworks in AI-driven learning environments remains sparse (Freiberg, 2023, pp. 62–68).

Additionally, despite the increasing emphasis on developing data literacy among educators, there is limited empirical evidence

connecting these competencies to AI adoption in classroom practices (Olari & Romeike, 2021).

An analytical approach was employed to examine the context of AI adoption among secondary school teachers in public schools in Catalonia. Thus, the purposive sampling method was used, as it is particularly effective in analysing complex educational settings where systemic, cultural, and policy-driven factors influence teacher behaviours. Given the emergent nature of AI adoption in education, an analytical approach (Salomon, 1991) in a specific context, such as Catalonia, allows for a context-sensitive analysis of how data literacy, AI knowledge, and sociodemographic variables shape teachers' engagement with AI-based tools. Furthermore, it complements the statistical analysis by offering rich empirical insights, facilitating a more nuanced interpretation of the PLS-SEM findings.

Considering the gaps detected, this article explores how these predictors—AI knowledge, perception, general data use, applied data use, and sociodemographic variables—contribute to AI adoption among Catalonia's secondary school teachers. Thus, the central research question of this case study is as follows.

- How do data literacy, AI knowledge, AI perceptions, and other sociodemographic variables influence secondary school teachers' use of AI technologies for educational purposes in Catalonia?

2.2. Sample

This work analyzed a purposive sample of 372 secondary school teachers in Catalonia, Spain. Given the study's focus on AI adoption in educational settings, a purposive sampling strategy (Tongco, 2007) was employed to ensure that participants were teaching in secondary education in Catalonia. The selection process followed an analytical approach (Salomon, 1991), which emphasizes examining how cognitive and contextual factors shape learning processes. Specifically, schools were included under the inclusion criteria of their institutional project, which explicitly described the structure and functioning of ICT within their educational framework. This ensured that the sampled institutions had a documented commitment to digital education.

Participants were recruited through institutional outreach, with prior approval from their high schools. Thus, the Snowball Sampling approach was developed to reach educators from each institution. As this method can introduce bias due to participants' tendency to recruit individuals with similar characteristics, key demographic factors that could influence AI adoption, such as teaching speciality, gender, years of experience, and age, were carefully controlled to enhance the validity of the findings (Marcus et al., 2017).

This sample size exceeded the minimum threshold of 160–200 observations required for reliable results in multivariate models with multiple predictors and small effect sizes (Kock & Hadaya, 2018), ensuring sufficient statistical power to use the PLS-SEM model.

According to ethical research standards, participation remained voluntary and anonymous. Thus, the items in the online survey were mandatory, ensuring that no data was missing.

As a result, the sample across these demographic variables minimized potential biases. The inclusion and exclusion criteria are developed as follows (Table 1).

2.3. Conceptual model

The proposed conceptual model is grounded in theoretical findings on AI adoption and educational data literacy. It comprises five key constructs: AI Perception, AI Knowledge, General Data Use, Applied Data Use, and STEM background, all of which are hypothesized to influence AI Adoption (Fig. 1).

Concretely, to operationalize these constructs, two validated instruments were employed, each specifically designed and validated for use among secondary education teachers.

Table 1
Sample's inclusion and exclusion criteria.

Inclusion Criteria	Exclusion Criteria
Public and/or publicly funded (concerted) educational institutions.	Private educational institutions.
High schools offering formal education within the Autonomous Community of Catalonia.	High schools located in autonomous communities outside Catalonia.
Secondary education institutions.	Early childhood, primary, and post-compulsory education institutions.
High schools whose institutional projects explicitly describe the functioning and structure of ICT in their educational project.	High schools without a documented ICT structure in their institutional project.

The Artificial Intelligence scales utilized in this study are derived from the questionnaire developed by Pallejà-Masip et al. (2024), which initially conceptualized three dimensions: AI Perception, AI Knowledge, and AI Usage. This study focuses on two of these dimensions—AI Perception and AI Knowledge—as predictors of AI Adoption. AI Perception was defined as concerns related to the use of AI, including issues of trust, transparency, and ethical implications. In contrast, AI Knowledge encompasses competencies associated with AI-driven content creation and application.

In the case of the predicted AI Adoption construct, the selected items were designed to capture a broad range of AI-related practices, thus ensuring sufficient variability and discriminatory power across different usage levels. The items included measure knowledge and frequency of use of AI tools and resources in teaching practice, as well as specific applications of Generative AI, such as image generation, programming code generation, lesson planning and preparation, and the design of assessment tasks (view Table 1). This structure allows for a nuanced understanding of how teachers integrate AI into their professional practice, accommodating both low and high levels of adoption.

The Data Literacy instrument employed was initially conceptualized by Donate-Beby et al. (2025) with three dimensions: Comprehensive Educational Analytics, Educational Problem-Solving Through Data, and Promoting Meta-Learning Students Through Data and Ethical Implications. However, following a Confirmatory Factor Analysis (CFA) conducted on the present sample, the final validated structure revealed two distinct dimensions that were used in this study: General Data Use and Applied Data Use. General Data Use reflects broader engagement with learning analytics practices (Siemens, 2013), whereas Applied Data Use encompasses more targeted, decision-oriented applications of educational data aimed at improving instruction.

According to the authors (Donate-Beby et al., 2025), the instrument explores categories of data commonly used in K–12 education, aligning them with the tasks and competencies typically emphasized at these stages. These categories include.

(1) **Student Behavioural Data:** Records related to students' attendance, classroom participation, homework submission, and other observable behaviors during the learning process. These data points allow educators to monitor engagement and adjust teaching strategies accordingly.
(2) **Performance and Assessment Data**: Information derived from formal assessments (such as exams, quizzes, and standardized tests) and informal evaluations (like classroom observations and student activities). This category supports the evaluation of student progress and the effectiveness of instructional practices.
(3) **Learning Analytics Data**: Data generated through digital learning environments, including metrics like click patterns, assignment submission timestamps, and time spent on tasks. Such data helps educators identify learning behaviours and optimize instructional methods based on student interaction patterns.

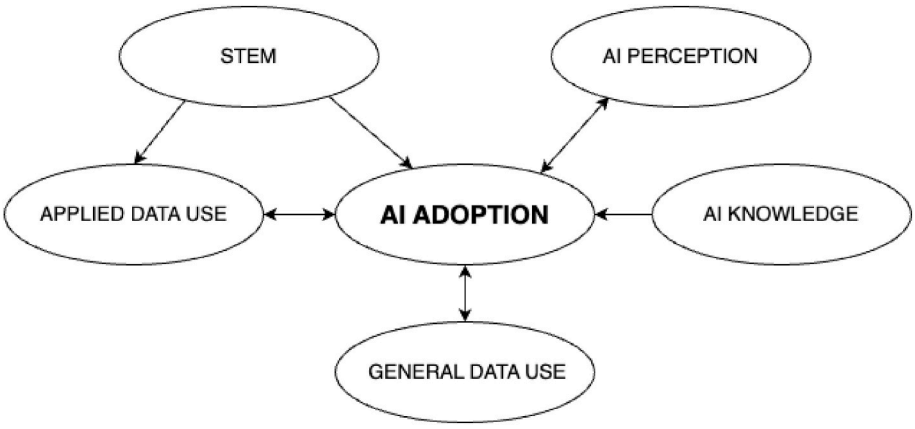


Fig. 1. Hypothesized relationships in the model.

(4) **Meta-Learning and Ethical Data**: Data that encourages critical reflection on the responsible management of sensitive information, highlighting issues related to privacy, data protection, and the ethical use of personal student information.

Due to the exploratory nature of this study, bidirectional relationships were examined among AI adoption, General Data Use, Applied Data Use, and AI Perception to capture a broader range of potential interactions.

The theoretical significance of AI adoption as the central phenomenon is evident, as studies consistently link various factors to its uptake (Choi et al., 2023; Sun et al., 2024; Li, 2024; Aguilar-Cruz & Salas-Pilco, 2025). General Data Use emerges as a component of broader technical mastery (Viberg et al., 2024), suggesting its potential influence on AI interaction. Applied Data Use, is implied through the concept of "AI Literacy" as the ability to utilize AI as a tool (Du et al., 2024), indicating the importance of practical data application within AI systems. Critically, AI Perception, encompassing perceived benefits, ease of use, and usefulness, consistently correlates with AI adoption (Aguilar-Cruz & Salas-Pilco, 2025; Choi et al., 2023; Du et al., 2024; Hwang & Chien, 2024; Li, 2024; Sun et al., 2024; Viberg et al., 2024), aligning with established models like TAM and highlighting its role in overcoming adoption hurdles related to ethical concerns and transparency (Aguilar-Cruz & Salas-Pilco, 2025).

Specifically, we have conceptualized AI Perception as a concern related to the use of AI, encompassing issues of trust, transparency, and ethical implications. Thus, AI Knowledge focuses specifically on competencies related to AI-driven content creation. At the same time, General Data Use represents a broad engagement with learning analytics practices (Siemens, 2013). In contrast, Applied Data Use involves targeted, decision-oriented engagement with data (view Table 1).

Additionally, professionals with STEM backgrounds typically demonstrate higher levels of computational thinking, problem-solving, and data literacy (Weintrop et al., 2016). Thus, the teaching specialization (STEM variable, view Table 1) was included in the hypothesized model to explore its influence on AI Adoption and Applied Data Practices, while other initially considered sociodemographic variables were excluded for model fit optimization.

2.4. Measurement model and variables

This study employed two validated instruments: one designed to examine constructs associated with Artificial Intelligence (AI) and the other to evaluate competencies in Data Literacy.

Before their integration into the Partial Least Squares Structural Equation Modelling (PLS-SEM) framework, Confirmatory Factor Analyzes (CFA) were conducted to validate the factorial structure of both instruments. Given the data's ordinal nature, the Diagonally Weighted Least Squares (DWLS) estimator was applied to ensure robust parameter estimation. This method was used because it is particularly suitable for ordinal scales, enabling robust estimations and effectively adjusting the model in the presence of skewed and kurtotic distributions (DiStefano & Morgan, 2014).

Although conducting a Confirmatory Factor Analysis (CFA) is not required before Partial Least Squares Structural Equation Modelling (PLS-SEM), where the measurement model is inherently assessed during the estimation process (Hair et al., 2022), a CFA was conducted in this study to enhance methodological transparency and provide preliminary evidence of construct dimensionality. In this study, we conducted a confirmatory factor analysis (CFA) to enhance methodological transparency and provide preliminary evidence of construct dimensionality. This section outlines the validation procedures undertaken before the structural model analysis.

The Data Literacy instrument was initially conceptualized with three dimensions: Comprehensive Educational Analytics, Educational Problem-Solving Through Data, and Promoting Meta-Learning Students Through Data and Ethical Implications (Donate-Beby et al., 2025). During the design phase, the authors implemented a comprehensive selection process that included a bibliographic review, expert judgment, and Exploratory Factor Analysis (EFA). The sample used to validate the questionnaire represented the country of origin and educational levels targeted in this study, ensuring that the items were understood and applicable. Thus, the Kaiser-Meyer-Olkin (KMO) measure indicated sampling adequacy (KMO = 0.905), and the Bartlett's Test of Sphericity confirmed the suitability of the data for factor analysis. Through Principal Component Analysis and Maximum Likelihood Extraction with Varimax rotation, a three-factor solution was retained, explaining 77.03 % of the total variance. The rotated factor matrix revealed clear and interpretable loadings above 0.60, supporting the proposed dimensions. Items that did not meet the established thresholds were systematically excluded, enhancing the construct validity. The final instrument configuration preserved 23 items distributed across the three dimensions, achieving excellent internal consistency (Cronbach's Alpha = 0.961 overall; Factor 1 = 0.976, Factor 2 = 0.945, Factor 3 = 0.963).

The Confirmatory Factor Analysis (CFA) results for the Data Literacy questionnaire finally revealed two dimensions: Applied Data Use and General Data Use. It showed an excellent overall fit for the hypothesized measurement model. Although the Chi-Square (χ^2/df = 0.000) suggested a saturated model due to perfect or near-perfect model specification, the comparative indices offered more substantial evidence. Specifically, the Comparative Fit Index (CFI = 0.997) and the Tucker–Lewis Index (TLI = 0.997) exceeded the recommended threshold of 0.95, indicating an excellent comparative fit relative to a null model. The Root Mean Square Error of Approximation (RMSEA = 0.071) fell

within the acceptable range (≤0.08), suggesting a reasonable approximation error. However, it was slightly above the threshold for close fit (0.06). Finally, the Standardized Root Mean Square Residual (SRMR = 0.036) was well below the 0.08 cut-off, supporting excellent absolute fit.

The standardized factor loadings (λ) were high and statistically significant ($p < .001$), with values ranging from 0.75 to 0.90, indicating that the items effectively represent their respective constructs. Additionally, the correlation between Applied Data Practices and General Data Use was 0.835 ($p < .001$), suggesting a strong positive relationship. These findings support the construct validity of the Data Literacy measurement model and provide a strong foundation for its integration into the subsequent PLS-SEM analysis.

The Artificial Intelligence scales utilized in this study are derived from the questionnaire design created by Pallejà-Masip et al. (2024). This design emphasizes three key dimensions: AI Perception, AI Knowledge, and AI Usage, specifically among pre-university teaching staff in Spain and Latin America. Following the results of this work, the instrument's development proceeded through a rigorous methodological process, which included a theoretical review of AI integration in education, the conceptual delimitation of key constructs, and validation of expert judgment. A panel of seven experts evaluated an initial tool of 64 items for content validity, following established procedures (Grant & Davis, 1997). Each item's clarity and relevance were assessed using a 4-point Likert scale, retaining items with at least 80 % expert agreement to ensure cultural adaptation. Subsequently, the instrument's internal consistency was evaluated through Cronbach's alpha (α = .946) and polychoric alpha (α = .943), which indicated excellent reliability. Split-half reliability testing further confirmed the questionnaire's robustness, with a Spearman-Brown corrected coefficient of 0.98. After the construct validity assessment using Bartlett's test of sphericity ($\chi^2 = 9574.17$, $p < .001$) and the Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy (KMO = 0.891), the Exploratory Factor Analysis (EFA) was developed, leading to the retention of three factors, consistent with the conceptualized initially structure.

Before the PLS-SEM, a Confirmatory Factor Analysis (CFA) was conducted to validate the factorial structure of the AI questionnaire before integrating it into the PLS-SEM model. The results of the CFA demonstrated an excellent model fit: $\chi^2/df = 0.980$, CFI = 1.000, TLI = 1.008, RMSEA = 0.000, and SRMR = 0.056. Specifically, the χ^2/df ratio was significantly below the recommended threshold of 3.0, while the CFI and TLI both exceeded 0.95. The RMSEA indicated a perfect fit at 0.000, and the SRMR was below the threshold of 0.08. Collectively, these indices confirm the adequacy of the measurement structure.

The standardized factor loadings (λ) ranged from 0.895 to 0.995, all statistically significant ($p < .001$), thereby supporting the convergent validity of the constructs (Hair et al., 2022). Specifically, the AI Knowledge dimension loadings ranged from 0.944 to 0.960, for AI Usage from 0.895 to 0.995, and for AI Perception from 0.929 to 0.968. All loadings exceeded the recommended 0.70 threshold, indicating strong measurement properties.

Furthermore, covariances among the three latent constructs were examined. A strong positive covariance was found between AI Knowledge and AI Usage (0.537). In contrast, moderate negative covariances were observed between AI Knowledge and AI Perception (−0.076) and between AI Usage and AI Perception (−0.152). These patterns are theoretically consistent with the notion that greater concerns about AI may inversely relate to its knowledge and usage among educators.

Collectively, the CFA results provide robust preliminary evidence of the AI instrument's construct validity and dimensionality, thus justifying its subsequent use within the PLS-SEM framework for structural model testing (Hair et al., 2022).

Initially, the demographic variables of Teaching Experience, Age, Gender, and Educational Role were tested as potential categorical predictors of AI adoption. The most parsimonious model was selected based on the Bayesian Information Criterion (BIC), which was used to compare different model configurations and assess whether modifications improved model fit (Neath & Cavanaugh, 2012).

Based on these comparisons, the most streamlined model was retained, excluding demographic variables while keeping the STEM specialization. Additionally, an alternative model was explored where AI Adoption was treated as an exogenous construct predicting AI Perception, Applied Data Use, and General Data Use. Hence, the hypothesized model is displayed in Fig. 1.

The final model specification, including AI Knowledge, AI Perception, General Data Use, Applied Data Use, and STEM Specialization as predictors of AI Adoption, is further analyzed in the *Results Section*. There, standardized path coefficients (β) and significance levels (p-values) were presented to assess the statistical relevance of each construct.

Thus, Table 1 summarizes the variables used in the model testing. Specifically, Table 1 presents each latent construct alongside its associated indicators (items) and its corresponding response scales, providing a comprehensive overview of the measurement model. The item identifiers have been standardized and are included to facilitate traceability and enhance the interpretability of the subsequent *Results Section.*

2.5. Data preprocessing and standardization

Before the PLS-SEM analysis, the dataset was subjected to a preprocessing phase to ensure consistency across all variables. Since all survey responses were mandatory, no missing data imputation was required. Data integrity was verified using SmartPLS 4, confirming the absence of system-missing values.

To ensure comparability across indicators, the ordinal scales for AI and Data Literacy were standardized using the Min-Max normalization method, transforming all ordinal variables into a range of 0–1.

Additionally, the outlier detection process was performed using data visualization techniques in R Studio. Given the exploratory nature of this study and the need to maintain generalizability, outliers were retained in the final dataset.

2.6. PLS-SEM model estimation and validation

The present work uses SmartPLS 4 to apply Partial Least Squares Structural Equation Modelling (PLS-SEM) to examine the relationships between AI adoption and its predictors (Hair et al., 2022).

A Partial Least Squares Structural Equation Model (PLS-SEM) consists of two interconnected sub-models: the measurement and structural models. In this study, the constructs were modeled using a reflective measurement approach, where the indicators are assumed to be manifestations of their respective latent variables. The measurement model evaluates the relationships between observed indicators and latent constructs, while the structural model examines the relationships among constructs within the model (Zhang & Wu, 2024). The assessment of the measurement model was conducted based on the following criteria.

- **Model fit** was assessed using the Standardized Root Mean Square Residual (SRMR), with values below 0.08, indicative of an acceptable fit (Henseler et al., 2016)
- **Reliability**: Composite reliability (CR > 0.7) and Cronbach's alpha (>0.7).
- **Convergent Validity**: Average Variance Extracted (AVE >0.5).
- **Discriminant Validity**: Fornell-Larcker criterion and HTMT ratio.

Moreover, the structural model was evaluated using the following criteria.

- **Bootstrapping procedure**: The bootstrapping procedure was further applied to test the significance of all total effects within the model using the *t*-test. It was conducted with 372 cases, using 5000 resamples and no-sign-changes options. The **two-tailed *t*-test** was

employed to assess the significance of the path coefficients in the inner model and determine the significance of the outer model loadings.

- **Collinearity Check**: Variance Inflation Factor (**VIF**) values were computed using construct scores as input for multiple regression analysis. A **VIF threshold of five** was used to ensure no significant multicollinearity issues (Hair et al., 2022).
- **Coefficient of Determination (R^2)**: Evaluates the model's explanatory power.
- **The effect size** (f^2): Determines the explanatory power of the endogenous constructs. Thus, values of 0.02 are considered small, 0.15 is medium, and 0.35 represents a significant effect of the exogenous constructs (Cohen, 1988).
- **The Normed Fit Index (NFI):** This assesses the model fit, with a maximum of 0.90 for an adequate model fit (Bentler & Bonett, 1980).
- **Predictive validity (Q^2, RMSE, and MAE):** This method estimates Q^2 Predict values, where values above zero confirm predictive validity, with thresholds of 0.02 (weak), 0.15 (moderate), and 0.35 (strong) (Hair & Alamer, 2022). A blindfolding procedure with an omission distance of D = 7 was used to compute cross-validated redundancy measures. This test was complemented with the Root Mean Square Error (RMSE) and the Mean Absolute Error (MAE) analysis. Thus, lower RMSE and MAE values indicate better predictive performance (Shmueli et al., 2019).

3. Results

3.1. Measurement model review

As Table 2 shows, the constructs met reliability and validity thresholds. Thus, the Heterotrait-monotrait ratio and Fornell-Lacker Matrix results confirm discriminant validity (Tables 3 and 4).

Table 4 presents the Fornell-Larcker discriminant validity matrix, where the diagonal values represent the square root of the average variance extracted (AVE). At the same time, the off-diagonal elements denote the correlation coefficients between constructs. The Fornell-Larcker criterion indicates discriminant validity when the square root of AVE for each construct exceeds its correlations with other constructs, which is the case (Fornell & Larcker, 1981). Thus, the correlations between constructs are low, suggesting that they capture distinct aspects of the measured phenomenon.

3.2. Structural model assessment

The final dataset comprised 32 indicators, structured into five exogenous constructs (predictors) and one endogenous construct (outcome), as displayed in Table 5.

Accordingly, the final model was selected based on its balance of predictive accuracy (R^2), model fit (SRMR), and parsimony (BIC), retaining AI Adoption as the endogenous variable and excluding demographic predictors (view Table 6). This selection optimizes the trade-off between explanatory power and model complexity. BIC favored the structure with AI Adoption as the dependent construct, SRMR indicated an acceptable model fit (0.051), and R^2 confirmed a reasonable proportion of explained variance (0.305).

Alternative model specifications resulted in a poorer model fit or excessive complexity (view Table 5), as evidenced by higher BIC values and weaker predictive contributions on AI adoption as an exogenous variable (Model 2) or by incorporating demographic variables into Model 1 (Model 4). Additionally, while Model 3 indicated a significant BIC, its estimation of Applied Data Use was unstable ($R^2 = 0.002$, $\Delta BIC = 10.058$), leading to its exclusion.

The statistical significance of path coefficients was assessed using bootstrapping with 5000 resamples in SmartPLS 4, following a percentile bootstrap approach with a two-tailed test ($\alpha = .05$). This non-parametric method does not assume normality and provides robust

Table 2
Latent constructs and indicators of the SEM.

Latent Variable	Identifiers	Scale	Item
AI Knowledge	AIK1	1.None 2.Low	Generate images with AI
	AIK2	3.Moderate 4.High	Generate videos with AI
	AIK3	5.Expert	Generate music with AI
AI Perception	AIP1	1.I am not worried at all 2.Somewhat	Reduction of fact and data verification
	AIP2	concerned 3.Moderately	Reduction of critical thinking
	AIP3	concerned 4.Very worried	Inaccuracies and errors
	AIP4	5.Extremely concerned	Bias and discrimination
	AIP5		Ethics and academic integrity, honesty, and misconduct (Plagiarism and cheating)
	AIP6		Privacy issues
	AIP7		Assessment of skills and knowledge
General Data Use	GDU1	1.I do not feel capable 2.I am capable of achieving this task with help. 3.I am capable of achieving this task in an elemental way. 4.I am capable of achieving this task entirely. 5.I am developing this task in my professional practice. 6.I am developing this task in my professional practice, and I can help and teach others.	To use data from digitized learning environments (e.g., virtual classroom learning activities, applications, or other tools) to identify the learning problems of the students and groups I teach.
	GDU2		To use data from digitized learning environments to solve problems in educational practice with the students and groups I teach.
	GDU3		To document problem-solving processes identified through data using reports, memos, journals, or other tools.
	GDU4		To use different data sources to get knowledge from my teaching practice and student learning.
	GDU5		To use evaluation, content transmission, or visualization tools that allow me to work with raw data to extract quick and elaborate results (e.g., Google/ Microsoft Forms, Moodle Quiz, Google Data Studio)
			To know privacy policies and legal implications of data management and disclosure in the educational classroom context.

(*continued on next page*)

Table 2 (*continued*)

Latent Variable	Identifiers	Scale	Item
Applied Data Use	ADU1 ADU2		To know the educational institution's organizational chart, roles, and internal processes related to data management and disclosure of data in the educational context (e.g., the institution's Data Protection Officer and the institution's Data Protection Guide).
	ADU3		Monitor students' performance through data generated in digital learning environments.
	ADU4		Generate actionable insights into students' performance and needs through the management of data generated in digital learning environments (e.g., using students' grades to generate insights into the causes of learning difficulties)
	ADU5		Understand the individual, social, and cultural learning contexts in which data has been collected for decision-making.
	ADU6		Encourage students to engage in self-learning, supervision, and progress tracking based on their data generated in digital learning environments (e.g., virtual classrooms and applications facilitating learning activities).
AI Adoption	AIA1	1.I am not aware of any AI tools or resources 2.I am aware of AI tools and resources, but I do not use them 3.I use AI tools and resources occasionally 4.I use AI tools and resources regularly 5.I use AI tools daily.	Knowledge and frequency of use of artificial intelligence tools and resources in teaching practice
	AIA2		Image generation
	AIA3	1.No experience at all 2.Limited experience 3.Some experience 4.Experienced 5.Highly experienced	Programming code generation
	AIA4		Plan/prepare lessons, activities, or learning experiences
	AIA5		Plan tasks and assessment
Sociodemographic variable	STEM	Categorical Variables (Yes/No)	**STEM:** Maths, Science, Technology **No STEM:** History, Special Education, Physical Education, Arts, Native Language, and Foreign Language
	Educational role		Headmaster/ headmistress, technical coordinator, pedagogical coordinator or teacher
	Teaching experience		0–5 years, 6–15 years, >15 years
	Age		<30 years, 30–39 years, 40–49 years, 50–59 years, <60 years

Table 3
Latent constructs' reliability and discriminant validity.

	Cronbach's alpha	Composite reliability (ρ_a)	Composite reliability (ρc)	Average variance extracted (AVE)
AI Adoption	0.904	0.909	0.929	0.724
AI_Knowledge	0.889	0.908	0.930	0.816
AI_Perception	0.913	0.944	0.930	0.655
Applied _Data Use	0.921	0.939	0.938	0.715
General _Data Use	0.927	0.934	0.944	0.773

Table 4
Heterotrait-monotrait ratio.

	Heterotrait-monotrait ratio (HTMT)
AI Knowledge <–> AI Adoption	0.525
AI Perception <–> AI Adoption	0.152
AI Perception <–> AI Knowledge	0.085
Applied Data Use <–> AI Adoption	0.268
Applied Data Use <–> AI Knowledge	0.252
Applied Data Use <–> AI Perception	0.077
General Data Use <–> AI Adoption	0.347
General Data Use <–> AI Knowledge	0.209
General Data Use <–> AI Perception	0.057
General Data Use <–> Applied Data Use	0.834

confidence intervals for model estimates. Path significance was determined using t-values, p-values, and 95 % confidence intervals, as presented in Table 7.

The path analysis shows no significant direct effect on Applied Data Use and STEM background ($\beta = -.063$, $p = .442$; $\beta = .063$, $p = .168$, respectively) over AI Adoption. In addition, AI Perception showed a weak but statistically significant negative path coefficient with AI Adoption ($\beta = -.105$, $p = .022$).

Conversely, the results indicate that AI Knowledge exhibited a statistically significant positive relationship with AI Adoption ($\beta = .430$, $p < .001$). General Data Use was also statistically significant ($\beta = .288$, $p = .001$).

Table 5
Discriminant validity – Fornell-Larcker matrix.

	AI adoption	AI knowledge	AI perception	Applied data use	General data use
AI adoption	0.851				
AI knowledge	0.481	**0.903**			
AI perception	−0.146	−0.079	**0.809**		
Applied data use	0.261	0.238	0.033	**0.846**	
General data use	0.324	0.192	−0.012	0.769	**0.879**

Table 6
Latent constructs and indicators of the SEM.

Construct	Type	Description
AI Knowledge	Exogenous	Familiarity with AI tools and concepts.
AI Perception		Teachers' attitudes and concerns regarding AI.
General Data Use		Basic data literacy usage in the teaching-learning process.
Applied Data Use		Practical applications of data literacy require advanced knowledge of data-driven activities.
STEM		Binary variable (1 = STEM, 0 = Non-STEM) assessing its influence on AI adoption.
AI Adoption	Endogenous	Self-reported use of AI tools in educational practices.

Table 7
Comparison of model specifications using Bayesian information criterion (BIC) and R.^{2}.

Model number	Model Specification	AI adoption BIC Value	R^2 (AI adoption)	R^2 (other variables)	Estimated SRMR
1	AI Adoption as an endogenous variable with STEM, AI Knowledge, AI Perception, General Data Use, and Applied Data Use as exogenous variables	−100.659	0.305	–	0.051
2	AI Adoption is an exogenous variable, while STEM, AI Knowledge, AI Perception, General Data Use, and Applied Data Use are endogenous variables.	−71.663	–	**AI knowledge** 0.232 **AI perception** 0.021 **Applied data use** 0.069 **General data use** 0.106 **STEM** 0.004	0.155
3	Model 1 with SEM acting as an exogenous variable of AI Adoption and Applied Data Practices.	**Model 3** −100.628 **Applied data use** 10.058	0.305	**Applied data use** 0.002	0.186
4	Model 1, including demographic variables	−83.425	0.312	–	0.048

The effect sizes (f^2) provide additional insight into the magnitude of these relationships (view Fig. 1). AI Knowledge exhibited a medium effect size *(f2 = .252)*, whereas General Data Use had a small but significant effect *(f2 = .048)*. In contrast, AI Perception *(f2 = .016)*, Applied Data Use *(f2 = .002)*, and STEM background *(f2 = .006)* had negligible effects, reinforcing the interpretation that these variables do not strongly influence AI Adoption.

3.3. Model fit and multicollinearity

The overall explanatory power of the model was assessed using R^2, which indicates that the predictors explain 30.5 % of the variance in AI Adoption ($R^2 = 0.305$), reflecting moderate predictive capability (Hair et al., 2022).

Multicollinearity was assessed using the Variance Inflation Factor (VIF). All values remained below the threshold of 5, confirming that collinearity was not a concern (Hair et al., 2022). The exact VIF values are reported in Appendix A.

In addition, the Standardized Root Mean Square Residual (SRMR) was examined, confirming an adequate model fit with a value below the acceptable threshold of 0.08 (Henseler et al., 2016). The Normed Fit Index (NFI = 0.861) falls below the commonly accepted threshold of 0.90. Although it is considered under the threshold for acceptable fit (>0.8) following other frameworks, the model fit could benefit from further refinement (Bentler & Bonett, 1980). While no absolute cutoff exists for the d_ULS (0.973) and d_G (0.485) indices, their lower values indicate that the estimated model does not substantially deviate from the observed data. Table 8 summarizes the model fit indices.

Moreover, the predictive relevance of the model was assessed using PLSpredict, a technique applied in educational research to evaluate out-of-sample predictive power (Hair & Alamer, 2022). The Q^2 Predict value for AI Adoption was 0.276, indicating moderate predictive validity. Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) were also examined to assess prediction accuracy. The relatively low RMSE (0.857) and MAE (0.641) suggested that the model provides reasonably accurate predictions (Shmueli et al., 2019).

The estimation results of the PLS-SEM model (Fig. 2) illustrate the structural relationships between AI Adoption and its hypothesized predictors, as indicated by the corresponding path coefficients (β). The model includes one observed categorical variable (STEM specialization) and four latent constructs—AI Knowledge, AI Perception, General Data Use, and Applied Data Use—each operationalized by multiple reflective indicators.

In Fig. 2, bold solid arrows indicate statistically significant and significant effects, with standardized path coefficients exceeding β > 0.40. Thinner solid arrows indicate statistically significant but moderate effects (β > 0.20), while dotted arrows illustrate negative relationships, and their thickness corresponds to the magnitude of the estimated coefficients.

Additionally, the model incorporates bivariate correlations between latent constructs (Table 9). These correlations complement the interpretation of the structural paths by revealing associations not directly modeled through causal links. Thus, Fig. 2 displays the correlations between the latent variables and AI adoption (see Table 10).

Standardized indicator loadings (λ) are included on each latent construct, indicating the extent to which each observed indicator reflects its underlying latent construct in the measurement model.

Furthermore, the coefficient of determination (R^2) for AI Adoption *(R2 = .305)* is displayed within the corresponding node in Fig. 2,

Table 8
Model path coefficients, mean, standard deviation, T-values, and P-values.

Path Coefficients (β)	Original Sample (O)	Sample mean (M)	Standard deviation (STDEV)	T Statistics	p- values	2.5 % CI	97.5 % CI	f^2
AI _Knowledge- > AI Adoption	**0.482**	0.430	0.045	9.528	0.00	−0.204	−0.033	***.252***
AI _Perception- > AI Adoption	−0.105	−0.116	0.046	2.287	0.022	−0.208	0.111	*.016*
Applied _Data Use - > AI Adoption	−0.063	−0.052	0.082	0.769	0.442	−0.028	0.150	*.002*
General _Data Use - > AI Adoption	**0.288**	0.282	0.090	3.186	0.001	0.104	0.462	***.048***
STEM - > AI Adoption	0.063	0.062	0.045	1.380	0.168	0.336	0.517	*.006*

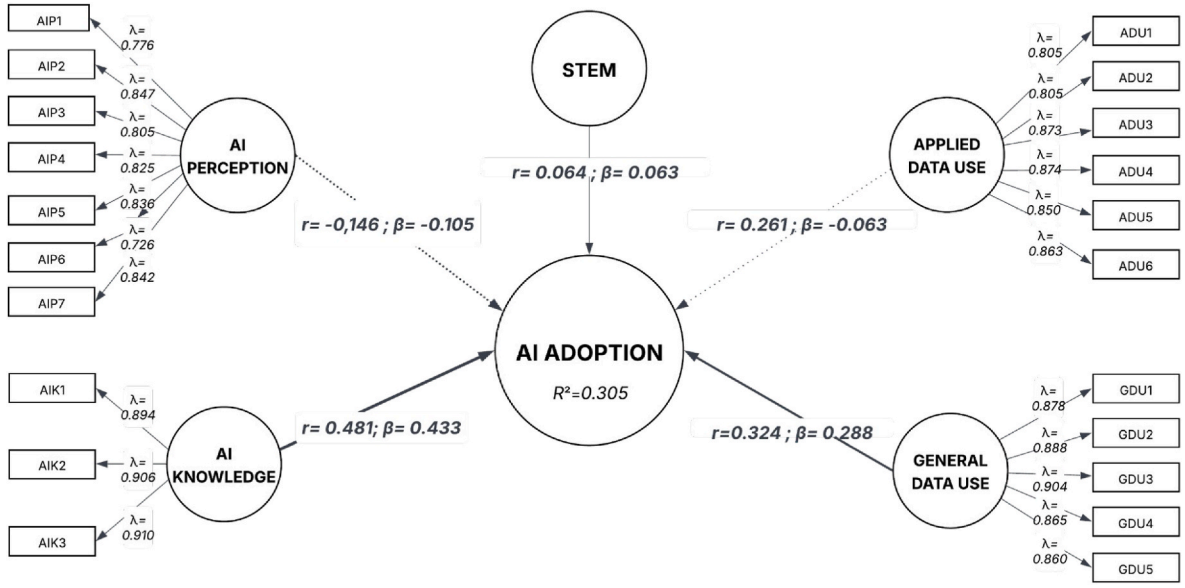


Fig. 2. PLS-SEM Final Graphical Output with the path coefficients (β), model R-square index, the indicator loadings on their latent variables (λ), and the correlations with AI Adoption.

Table 9
Model fit.

	Saturated model	Estimated model
SRMR	0.051	0.051
d_ULS	0.973	0.973
d_G	0.485	0.485
χ^2	1040.854	1040.854
NFI	0.861	0.861

indicating that approximately 30.5 % of the variance in AI Adoption is explained jointly by the predictors included in the model.

4. Discussion

Integrating Artificial Intelligence (AI) into teaching practices is a complex transformation that requires a deep understanding of various factors, from teacher perceptions to the competencies needed for effective implementation (Cui et al., 2023).

The methodology used in this study has already been used in prior research, as in the work of Frawley and Campbell (2025), which applied PLS-SEM to examine K-12 teachers' intentions to adopt emerging technologies. With a similar approach to the present work, they explored how factors, such as the attitude towards technology and the implicit rules, are the most notable antecedents in predicting the intention of use. However, rather than integrating a theoretical framework, as the Decomposed Theory of Planned Behaviour (DTPB), into a single predictive model, this study focuses on how competencies-related and AI perceptions interact with AI Adoption, acknowledging that technological factors often require context-specific analyzes.

Specifically, this work adopts a self-efficacy approach, recognizing that self-efficacy competence is a relevant indicator for describing latent variables and predicting behaviour (Holden & Rada, 2011). In the context of technology adoption, as mentioned Wang and Chuang (2024), perceived self-efficacy has been identified as a key factor in the acceptance and use of new information technologies. The greater a person's confidence in their ability to use a specific application, the greater the perceived usefulness of that application (Mendez-Carbajo, 2020). Thus, teachers' particular confidence in their abilities to understand, use, and integrate AI in education is a determining factor in their decision to adopt these technologies and in the way they implement them.

As noted in previous research, AI cannot be effectively integrated

Table 10
Correlations between latent variables.

	AI Adoption	AI Knowledge	AI Perception	Applied Data Use	General Data Use	STEM
AI Adoption	1					
AI Knowledge	0.481	1				
AI Perception	−0.146	−0.079	1			
Applied Data Use	0.261	0.238	0.033	1		
General Data Use	0.324	0.192	−0.012	0.769	1	
STEM	0.064	−0.002	−0.025	0.045	0009	1

into education without a solid foundation in data literacy (Olari & Romeike, 2021). In contrast, the relationship between data literacy and AI competencies has not yet been holistically examined through a statistical modelling approach until this work (Casal-Otero et al., 2023). Thus, the present article further addresses a research gap by employing Partial Least Squares Structural Equation Modelling (PLS-SEM) to analyze their interdependencies.

The final model explained 30.5 % of the variance in AI Adoption ($R^2 = 0.305$), with predictive relevance confirmed by $Q^2 = 0.276$ and an acceptable model fit (SRMR = 0.051). Additionally, RMSE (0.857) and MAE (0.641) indicated reasonably accurate predictive performance (Shmueli et al., 2019. Further, the model fit was adequate (SRMR <0.08), with a moderately strong Normed Fit Index (NFI = 0.861) (Bentler & Bonett, 1980).

AI is conceived as a tool capable of enhancing teacher productivity and supporting instructional tasks (Kehoe, 2023). All the dimensions evaluated in this work were derived from two questionnaires, and their validity and reliability were confirmed through the CFA's indices. Thus, AI adoption reflects the sample's current AI practices. As this study demonstrates, rather than analysing existing data, AI adoption involves actively leveraging Generative AI for content and data creation, such as producing images, lesson plans, activities, learning experiences, or programming code (Table 1).

The PLS-SEM analysis revealed that AI knowledge *($\beta = .482$, $p < .001$)* and General data use *($\beta = .288$, $p = .001$)* exhibited the most substantial path coefficients in predicting AI adoption. AI knowledge had a medium effect size *($f2 = .252$)*. In contrast, General data use had a small but significant effect *($f2 = .048$)*.

On the one hand, the AI knowledge variable, defined by generating AI-driven content such as images, videos, and music, significantly predicts AI adoption. Prior research has suggested that developing AI understanding facilitates more profound engagement with AI tools (Zhai et al., 2021), reinforcing that teachers who actively create with AI are more likely to integrate it into their instructional practices.

On the other hand, the General data practices variable aligns with learning analytics, as it embraces teachers' competencies in identifying how data can be analyzed to improve their teaching practices and students' learning (Siemens, 2013). Teachers can identify learning challenges, optimize instruction, and document data-driven decisions by leveraging digitized learning environments and data visualization tools (e.g., Google/Microsoft Forms, Moodle Quiz, and Google Data Studio).

Teachers interact with data using AI, as the teaching-learning process inherently involves using data in digitized environments (Gouseti et al., 2023). AI-driven teaching models have been developed based on learning data, providing objective feedback that supports reflective teaching practices (Hu, 2021; Li & Su, 2020). Consequently, competencies in General data practices are essential for AI Adoption, as they bridge data literacy with Generative AI applications.

Therefore, General data practices contribute to AI adoption by helping educators determine what data is relevant when prompting AI tools and effectively interpreting AI-generated outputs. Further, a teacher trained in General data practices may analyze student performance data and use this insight to refine AI-generated instructional materials. For instance, they could effectively prompt Generative AI tools to create personalized exercises and critically evaluate the input and output quality (Asrifan et al., 2025).

In contrast, the Applied data use dimension, which assesses more technical aspects of data management (Table 1)—including ethical considerations in data visualization, privacy policies, legal implications, and institutional data governance—did not directly affect AI Adoption. These results indicate that Applied Data Use does not have a significant direct effect on AI Adoption *($\beta = -.063$, $p = .442$)*, with a negligible effect size *($f2 = .002$)*, suggesting that technical data management competencies alone do not drive AI adoption in educational contexts.

The STEM variable *($\beta = .063$, $p = .168$)* had a similar effect on AI adoption, with a small effect size *($f2 = .006$)*. This is expected since STEM competencies are closely connected to the development of computational thinking, problem-solving, and technical data literacy skills (Weintrop et al., 2016).

Previous research on AI perception, which encompasses concerns about AI implementation, has demonstrated that perceived risk moderates perceived ease of use (Shen & Chiou, 2010) and, consequently, may influence AI adoption. These findings are consistent with the results obtained in the model, where AI Perception exhibited a weak but statistically significant negative relationship with AI Adoption *($\beta = -.105$, $p = .022$)*. Although the effect size *($f2 = .016$)* indicated a minor influence, these results highlighted that concerns about AI play a role in AI usage, albeit with limited practical significance.

Overall, beyond AI Knowledge and General Data Practices as the strongest predictors, the results highlight a limitation when determining how STEM background, Applied Data Use, and AI Perceptions individually affect AI Adoption. Accordingly, these variables may contribute under specific conditions that warrant further investigation.

5. Conclusions and implications for education

Integrating Artificial Intelligence (AI) into teaching practices is a complex transformation that requires a deep understanding of various factors, from teacher perceptions to the competencies needed for effective implementation. This study has highlighted several key findings that contribute to our understanding of AI adoption among secondary education teachers in Catalonia.

Firstly, AI knowledge, particularly in content creation, emerged as a significant predictor of AI adoption. Teachers who are proficient in generating AI-driven content, such as images, videos, and music, are more likely to integrate AI into their instructional practices. This underscores the importance of providing targeted professional development that enhances teachers' AI content creation skills.

Secondly, general data practices aligned with learning analytics play a crucial role in AI adoption. Teachers' ability to use data from digitized learning environments to identify and address educational problems is essential for effective AI integration. This finding suggests that fostering data literacy among teachers should be a priority, as it bridges the gap between data-driven decision-making and AI applications.

However, the study also revealed that technical data management skills, represented by applied data use, do not significantly impact AI adoption. This indicates that while technical competencies are important, they alone are insufficient to drive AI integration in educational contexts. Instead, a focus on pedagogical applications of AI and data literacy could be more beneficial.

Conversely, teachers' STEM background showed a limited effect on AI adoption. While STEM competencies are closely related to computational thinking and problem-solving, they do not necessarily explain higher AI adoption rates. This finding highlights the need for interdisciplinary approaches that integrate AI literacy across diverse teaching specializations.

Concerns about AI implementation, captured by the AI perception variable, were found to have a weak but statistically significant negative relationship with AI adoption. Although the effect size was negligible, addressing these concerns through transparent communication and ethical guidelines remains essential to build trust and confidence among teachers.

6. Limitations and future research

The Partial Least Squares Structural Equation Modelling (PLS-SEM) approach (Hair et al., 2022) is particularly relevant for predictive research, prioritizing explained variance (R^2) and indicator reliability over rigorous model fit criteria. Unlike covariance-based SEM (CB-SEM), which emphasizes goodness-of-fit indices, PLS-SEM focuses on maximizing the explanatory power of latent constructs through path modelling. Hence, although PLS-SEM does not rely on absolute fit

indices such as CFI or RMSEA, it incorporates alternative measures such as SRMR, R^2, Q^2, and AVE to assess model quality and predictive relevance. This implies that PLS-SEM primarily establishes associative relationships rather than causal effects, suggesting that the results should be interpreted cautiously.

While the study's replicability is high due to transparency in data collection and reporting, its generalizability is constrained by the national context of the sample and the educational level examined. Despite continuous efforts to standardize European educational practices, such as the Digital Competence Framework for Citizens (Vuorikari et al., 2022), AI adoption trends and systemic policies still vary significantly across countries. Consequently, cross-national comparisons could extend the present findings as AI literacy and data-driven practices are embedded within broader institutional and policy contexts. Additionally, several limitations inherent to this research design must also be acknowledged.

First, the analysis of AI perceptions focused exclusively on concerns, excluding positive dimensions such as optimism or perceived usefulness. This decision was guided by the principle of parsimony to maintain a manageable model structure suitable for exploratory research. However, to contribute to a more comprehensive understanding of AI integration in education, we encourage future research to further investigate these contextual elements and broader perceptual dimensions.

Second, although the model demonstrates moderate explanatory power, this level is acceptable for studies employing PLS-SEM in early-stage theory development (Hair et al., 2022). Thus, while acknowledging the moderate explanatory power, this study lays a transparent foundation for future empirical efforts, encouraging the refinement and extension of the proposed model in diverse educational settings.

Third, reliance on self-reported measures introduces potential biases, a standard limitation even in widely used frameworks such as the Technology Acceptance Model (TAM; Wang et al., 2021), where constructs like perceived usefulness and perceived ease of use are based on subjective internal states. Nevertheless, such models and instruments provide valuable foundations for developing institutional policies and guiding future research initiatives.

Fourth, given the study's cross-sectional nature, the findings capture a specific moment in time and do not account for dynamic changes in AI adoption trends. In this line, longitudinal designs could address this limitation in future research.

Finally, the instruments used may not fully reflect the full diversity of emerging AI tools. Nevertheless, the instruments selected were specifically aligned with the practical realities of secondary education teachers, and their validation process was explicitly oriented toward this target population. Furthermore, since this study primarily focused on individual AI and Data Literacy competencies and perceptions, future research should expand the model by exploring external factors, such as institutional policies, technological infrastructure, and organizational support (Bauschard & Quidwai, 2024).

CRediT authorship contribution statement

Marta López-Costa: Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Belén Donate-Beby:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation. **Natividad Cabrera-Lanzo:** Writing – review & editing, Validation, Investigation, Conceptualization. **Marcelo Fabián Maina:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Investigation, Conceptualization.

7. Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Gemini in order to proof edit the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

8. Funding

This research was made possible through generous funding from the Departament d'Educació (Ref. No. EDU145/23/000032) of the Generalitat de Catalunya.

Declaration of competing interest

The authors declare that they have no competing interests.

Appendix A. Variance Inflation Factor (VIF)

Appendix 1. Variance Inflation Factor (VIF)

VARIABLES	VIF	VARIABLES	VIF
AIK1	2.044	**ADU1**	2.559
AIK2	3.343	**ADU2**	2.718
AIK3	3.298	**ADU3**	3.381
AIA1	2.732	**ADU4**	3.125
AIA2	2.725	**ADU5**	2.480
AIA3	2.544	**ADU6**	2.574
AIA4	3.079	**GDU1**	2.970
AIA5	1.838	**GDU2**	3.256
AIP1	2.243	**GDU3**	3.586
AIP2	2.751	**GDU4**	2.814
AIP3	3.090	**GDU5**	2.344
AIP4	3.092	**STEM**	1.000
AIP5	2.646		
AIP6	1.937		
AIP7	2.329		
INTERACTION	**VIF**	**INTERACTION**	**VIF**
AI Knowledge - > AI Adoption	1.069	*Applied Data Use - > AI Adoption*	2.524
AI Perception - > AI Adoption	1.013	*General Data Use - > AI Adoption*	2.460
		STEM - > AI Adoption	1.005

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